

Linear Programming Course Summary

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1 Key Terms

- **Slack Variable:** A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. The slack variable represents the difference between the left and right sides of the inequality.
- **Feasible Region:** The set of all points that satisfy all the constraints of a linear programming problem.
- **Vertex:** A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.
- **Basic Variable:** In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. Basic variables are set to values that satisfy the constraints.
- **Non-basic Variable:** In the Simplex method, a variable that is set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.
- **Pivot Operation:** The process of exchanging a basic and a non-basic variable in the Simplex method, leading to a new dictionary and a movement to a different vertex in the feasible region.
- **Dictionary:** A representation of a linear programming problem in a tabular format, where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move towards the optimal solution.
- **Auxiliary Problem:** A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.
- **Largest-Coefficient Rule:** A pivot rule in the simplex method that selects the entering variable with the largest coefficient in the objective function row of the dictionary.
- **Largest-Increase Rule:** A pivot rule in the simplex method that selects the entering variable that produces the largest increase in the objective function value.
- **Klee-Minty Example:** A worst-case example for the simplex method, demonstrating exponential time complexity for certain pivot rules.
- **Bland's Rule:** A pivot rule in the simplex method designed to avoid cycling, but not necessarily to improve efficiency.
- **Lexicographic Method:** A method for choosing the leaving variable in the simplex method to prevent cycling.
- **Smoothed Complexity:** A measure of the complexity of an algorithm that considers the average-case performance after small random perturbations to the input.
- **Interior Method:** A class of algorithms for linear programming that traverse the interior of the feasible region, in contrast to the simplex method which follows the boundary.

2 Problem Types

- **Finding an Initial Feasible Solution:** The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible starting point.
- **Handling Infeasible Origins:** The standard Simplex method assumes the origin is a feasible solution. Techniques are needed to handle cases where the origin is infeasible.
- **Optimizing in Higher Dimensions:** The Simplex method is demonstrated for problems with three variables, requiring a systematic approach to find the optimal solution.
- **Using Dictionaries to Represent Solutions:** The Simplex method uses dictionaries to represent the current solution and to guide the iterative process of moving from one vertex to another.
- **Analyzing the Efficiency of the Simplex Method:** This problem involves analyzing the average-case and worst-case performance of the Simplex method, considering factors like the number of iterations and the influence of different pivot rules. Tables 4.1 and 4.2 provide empirical data on the average number of iterations for different rules and problem sizes.
- **Understanding Worst-Case Behavior of the Simplex Method:** This problem focuses on understanding why the Simplex method can exhibit exponential time complexity in the worst case. The Klee-Minty example is used to illustrate this behavior, showing how specific problem structures can lead to a large number of iterations.
- **Evaluating Different Pivot Rules:** This problem involves comparing the performance of different pivot rules for the Simplex method, such as the largest-coefficient rule and the largest-increase rule. The goal is to determine which rule is more efficient in terms of both the number of iterations and the overall computation time.
- **Empirical Analysis of Simplex Method Performance:** This problem involves analyzing empirical data from various experiments to understand the practical performance of the Simplex method. A log-log plot is used to visualize the relationship between problem size and the number of iterations, revealing a power-law relationship.
- **Theoretical Bounds on Simplex Method Performance:** This problem explores theoretical results concerning the number of iterations required by the Simplex method. It discusses the difficulty of proving polynomial bounds for various pivot rules and introduces a randomized pivot rule with a proven sub-exponential bound on the expected number of iterations.
- **The Existence of Polynomial-Time Algorithms for Linear Programming:** This problem highlights the existence of polynomial-time algorithms for linear programming, such as interior-point methods, which contrast with the Simplex method's potential for exponential time complexity in the worst case.

3 Algorithm Solutions

- **Simplex Method:** Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function. This is equivalent to moving from vertex to vertex in the feasible region.
- **Pivot Operation:** Interchange one basic and one non-basic variable in the dictionary by rewriting the equations to express the basic variables in terms of the non-basic variables and updating the objective function accordingly.
- **Auxiliary Problem:** Introduce a new variable x_0 and modify the constraints to create a new problem whose feasible region includes the origin. Solve this problem to find a feasible solution for the original problem.

- **Klee-Minty Example:** Maximize $\sum_{j=1}^n 2^{n-j} x_j$ subject to $\sum_{j=1}^n 2^{i-j} x_j + x_i \leq 100^{i-1}$, $i = 1, 2, \dots, n$, and $x_j \geq 0$, $j = 1, 2, \dots, n$.
- **Largest-Coefficient Rule:** Select the non-basic variable with the largest coefficient in the objective function row of the dictionary to enter the basis.
- **Largest-Increase Rule:** Select the non-basic variable whose entrance into the basis yields the largest increase in the objective function value.
- **Bland's Rule:** Choose the entering variable with the smallest index among those with positive reduced costs. Choose the leaving variable with the smallest index among those that satisfy the minimum ratio test.
- **Lexicographic Method:** Resolve degeneracy by using a lexicographic ordering of the basic variables.
- **Smoothed Complexity:** If we take an arbitrary linear program and then change its coefficients by small random amounts, the simplex method with a certain pivot rule reaches the optimum in polynomially many steps with high probability.
- **Interior Method:** A polynomial-time algorithm for linear programming that traverses the interior of the feasible region, unlike the Simplex method which follows the boundary.